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INDUSTRY EXPERIENCE:

Novartis Pharmaceutical Corporation
East Hanover, NJ

Sr. Global Medical Director, Oncology Medical Affairs **2013 –Present**

- Implements an aligned Medical Affairs perspective on compound development and clinical trial designs
- Works collaboratively across functions to execute an integrated medical plan and strategies
- Manages all operational aspects and drives execution for assigned projects of clinical trials in partnership with global line functions and regional/country medical associates
- Provides medical leadership for Novartis sponsored global medical affairs trials
- Manages Phase 3/4 global clinical trials and investigator-sponsored trials
- Develops medical launch strategy and supports commercial activities in alignment with the Global Brand Team
- Provides medical support for scientific symposium and medical congresses within legal and compliance regulations
- Provides medical input on overall publication strategy and is a key member of publication planning team providing global medical affairs input on individual publication plans
- Collaborates across functions to achieve approval of key scientific communications and medical information documents within legal and compliance regulations
- Provides medical leadership in interactions with key external stakeholders
- Responsible to efficiently manage budgets and resources
- Ensures career development of direct and indirect reports through active participation in the performance management and talent planning processes
- Responsible for developing clinical components of key documents – regulatory & safety – from global medical affairs studies supporting submissions and post-marketing commitments to HA's

Sanofi-Aventis
Bridgewater, NJ **2005-2013**

Global Clinical Leader
2009 –2013

- Leads clinical-commercial optimization of product development (Phase 2-3) and life cycle management (LCM/Phase 3B/4) strategies
- Directs cross-functional teams of medical professionals (health outcomes, biostatistics, medical communications, safety, regulatory, and field liaisons) in planning and execution of all aspects of Medical Affairs product support
- Creates the US and Global Medical Plan Strategy to prioritize regional needs
- Partners closely with other TAs and product leaders to develop integrated Global Medical Affairs insights, opportunities, perspectives, strategies and plans

- Serves as an ad hoc member of the clinical teams, Global Marketing Teams and licensing/due diligence committees
- Accounts for the assigned compounds' publication plans, opinion leader relations, medical education (including symposia, speaker trainings) and global advisory boards for products starting in early development or proof of pharmacology/proof of concept (Phase 2A/2B) throughout launch of major LCM initiatives
- Ensures clinical input and integration with the overall global compound strategy for developing a Target Product Profile
- Reviews and approves Investigator Sponsored Trials (concepts/protocols) according to applicable SOPs

Director of Clinical Research **2005 – 2009**

- Responsible for Global Clinical Development of Phase 2 and 3 including clinical development plans (CDPs), protocols, informed consent forms, clinical study reports, budgets, and regulatory submissions
- Organized, managed and directed all activities of the clinical teams to complete the clinical development projects to submission (NDA and/or MAA) for several compounds
- Provided leadership for training programs in compounds, protocols and studied disease
- Informed management of clinical team activities through written reports and review meetings
- Networked with investigators, KOLs/KTLs, IDMC Chair and Members during the clinical studies
- Accurately interpreted and communicated clinical trial results
- Assured timely completion and quality and accuracy of study reports and manuscripts
- Actively participated in the development of SOPs, policies, departmental standards and other process improvement efforts
- Supported Regulatory teams submitting INDs, NDAs

2004 - 2005
Johnson & Johnson (Research Pharmaceutical Services)
Raritan, NJ

MD Oncology Consultant

- Contributed clinical and preclinical input to regulatory filing of NDAs in Oncology and Hematology

2003 - 2005
Medical Discovery Consultants

Consultant for pharmaceutical and biotech companies

- Research protocols (design and writing for Phase 2 and 3 (from IND to NDA) based on CDP and from launch to full commercialization (Phase 4, 3B/SNDAs)
- Input/review for regulatory, safety and marketing for biotech companies and CROs
- Case Report Forms – review and input for CDM (sponsor and CRO)
- Clinical Development Plans – design, input, writing and review
- Investigators Brochure – review, input, writing sections
- Responses to FDA letters/communication - clinical consult to Regulatory activities
- Safety reports - review and build integrated summaries of safety (dossiers) and Annual Reports
- Medical input to regulatory filing in oncology, hematology and inflammatory diseases

1994 - 2002
Carbomed, Inc.
Franklin, Tennessee

Group Leader (while on Faculty at Vanderbilt University)

- Achieved the integration of early development programs with drug discovery in terms of
  - Disease indications
  - Identification of biomarkers
  - Segmentation of patient populations
  - Use of genomic approaches in proposed clinical development
- Created pre-submission documents including preclinical/tox data and clinical/safety reports (dossier)
- Presented preclinical and clinical data at meetings with FDA and other health authorities
- Designed Phase 2A (Proof-of-Pharmacology/Proof of Principle) studies in Oncology and CNS trauma programs
- Designed of Proof-of-Concept (Phase 2B) and pivotal studies (Phase 3 registration programs) and protocols
- Created Clinical Study Reports for Oncology and Autoimmune disorder studies
- Wrote and reviewed manuscripts, posters, abstracts, reports and patent applications

ACADEMIC and HOSPITAL/CLINICAL EXPERIENCE

1996-2002 **Assistant Professor**
Department of Biochemistry
Vanderbilt University School of Medicine, Nashville, TN
(1994-2002 Group Leader and other positions at Carbomed, Inc. while on faculty at Vanderbilt University School of Medicine)

1994-1996 **Research Instructor in Biochemistry**
Department of Biochemistry
Vanderbilt University School of Medicine, Nashville, TN

1992-1994 **Instructor in Obstetrics and Gynecology, Research**
Department of Obstetrics and Gynecology
Vanderbilt University School of Medicine, Nashville, TN

1989-1992 **Clinical and Research Fellow**
Department of Obstetrics and Gynecology
The Vanderbilt Hospital, Nashville, TN

1987-1989 **Attending Physician**
Practice in General Medicine and Pediatrics
Hospitals of Zespol Opieki Zdrowotnej, Lublin, Poland

1987-1993 **Clinical Sub-Investigator**

- Phase 3-4 clinical trials while at Medical Academy Hospitals, Lublin and Vanderbilt University Medical Center, Nashville, TN
- Open-label extensions
- Investigator Initiated Clinical Research

EDUCATION AND CLINICAL EXPERIENCE:

1989-1992 **Clinical & Research Fellow in OB/GYN and Reproductive Medicine**
Department of Obstetrics & Gynecology
and Center for Fertilization and Reproductive Research
Vanderbilt University Medical Center, and
The Vanderbilt University Hospital, Nashville, TN

1988-1989 **Resident Physician** (PGY-2), Pediatrics-Neonatology (relocated to the United States with family during the 2nd postgraduate year of training.
Completed GME trainings in Internal Medicine (PGY-3 and 4) in 1994 and was awarded a Certificate of General Practitioner (**G.P. in Primary Care Medicine**)

1987-1988 **Resident Physician** (PGY-1), Rotating internship (3 months each rotation) in Medicine, Surgery, Pediatrics, and Obstetrics & Gynecology.
Lublin University Hospitals and Clinics, Institute of Pediatrics, Institute of Internal Medicine, Institute of Surgery, Institute of OB/GYN, Medical Academy of Lublin, Poland

1980-1987 **Medical Student**, Lublin University School of Medicine, Lublin, Poland
(6-year M.B.B.Ch. at World Health Organization (WHO) accredited program) **M.D. 1987**

LICENSURE AND CERTIFICATION:

1987 **Doctor of Medicine** (M.D.) - Diploma

1987 **Unrestricted License to Practice in Medical Profession** - Certificate, Department of Health, Lublin, Poland

1994 **Primary Care Medicine** - Certificate (Medical Doctor in Primary Care and Family Medicine with Pediatric and Obstetric privileges as General Practitioner; **GP**) Council of Physicians of Poland, General Chamber of Physicians, Surgeons and Dentists

1994 **MD License** (achieved credentials for EU General Practice License as Attending Physician with unrestricted hospital privileges)

2004 **GP/Primary Care License** (re-certified for EU effective May 1st 2004)

HOSPITAL APPOINTMENTS:

1989-1993 **Clinical and Research Fellow in Reproductive Medicine**
The Vanderbilt Hospital
Department of Obstetrics & Gynecology
Vanderbilt University Medical Center
Nashville, TN

1987-1988 Staff Physician
Hospital of Zespol Opieki Zdrowotnej
Department of Pediatrics
Division of Neonatology, Lublin, Poland

1987-1989 Attending Physician
Hospital of Zespol Opieki Zdrowotnej

Department of Emergency Medicine
Emergency Room, OB ER, and Pediatric ER

1987-1993 Clinical Sub-Investigator/Investigator: Phase 2-3-4 clinical trials at Medical Academy Hospitals and The Vanderbilt Hospital
Randomized Controlled studies, Open-label studies and extensions, Investigator Initiated Research Studies (IIRPs, Private INDs)

BIBLIOGRAPHY:

1. Ian Smith, Denise Yardley, Howard Burris, Richard De Boer, Dino Amadori, Kristi McIntyre, Bent Ejlertsen, Michael Gnant, Walter Jonat, Kathy Pritchard, Mitch Dowsett, Lowell Hart, Susan Poggio, Lisa Comarella, Herve Salomon, Barbara Wamil, Joyce O'Shaughnessy. Comparative Efficacy and Safety of Adjuvant Letrozole vs Anastrozole in Postmenopausal Patients With Hormone Receptor-Positive, Node-Positive Early Breast Cancer: Final Results of the Randomized Phase 3, Femara vs Anastrozole Clinical Evaluation (FACE) Trial. J Clin Oncology 2017; 35(10):1041-1048

2. Mark S. Freedman, Jerry S. Wolinsky, Barbara Wamil, Christian Confavreux, Giancarlo Comi, Ludwig Kappos, Tomas Olsson, Aaron E. Miller, Hadj Benzerdjeb, Huiling Li, Catherine Simonson, and Paul O'Connor. Teriflunomide added to interferon-beta in relapsing multiple sclerosis: a randomized phase II trial. Neurology 2012; 78(23):1877–1885

3. P. O'Connor, J. Wolinsky, C. Confavreux, G. Comi, L. Kappos, T.P. Olsson, H. Benzerdjeb, B. Wamil, L. Wang, A. Miller, M.S. Freedman for the Teriflunomide Multiple Sclerosis Trial Group. A placebo-controlled phase III trial (TEMSO) of oral teriflunomide in relapsing multiple sclerosis: clinical efficacy and safety outcomes. 2010 ECTRIMS (European Committee for Treatment and Research in Multiple Sclerosis), Mult scler; 2010; 16: S7-S39.

4. Ohlsson M., Mattsson P., Wamil B.D., Hellerqvist C.G and Svensson M.: Macrophage stimulation using a group B-streptococcus exotoxin (CM101) leads to axonal regrowth in the injured optic nerve. Restor. Neurol. Neurosci. 2004; 22(1):33-41

5. Issued Patent: "Facilitation of wound healing with CM101/GBS toxin". Hellerqvist; C.G, Neeman M., Wamil B. D., Abramovitch R: U.S. Patent No. 6,670,337 - Granted December 30, 2003.

6. Issued Patent:" Facilitation of keloid healing with CM101/GBS toxin". Hellerqvist; C.G. Neeman M., Wamil B., Abramovitch R: U.S. Patent No. 6,569,838 - Granted May 27, 2003.

7. Issued Patent:"Treatment of Chronic Inflammatory Diseases with CM101/GBS Toxin". C.G. Hellerqvist C.G. and Wamil B.D: U.S. Patent No. 6,476,002 - Granted November 5, 2002.

8. Issued Patent: "Facilitation of repair of neural injury with CM101/GBS toxin. Hellerqvist C. G, Wamil; A.W, Wamil; B.D: U.S. Patent No. 6,476,001- Granted November 5, 2002.

9. Fu C., Bardhan S., Cetateanu N.D., Wamil B.D., Wang Y., Yan H-P., Shi E., Carter C., Venkov C., Yakes F.M., Page D.L., Lloyd R.S., Mernaugh R.L., Hellerqvist C.G.: Identification of a novel membrane protein, HP59, with therapeutic potential as a target of tumor angiogenesis. Clin Cancer Res. 7(12):4182-94, 2001.

10. Nanney L.B., Wamil B.D., Whitsitt J., Cardwell N.L., Davidson J.M., Yan H-P., and Hellerqvist C.G.: CM101 stimulates cutaneous wound healing through an anti-angiogenic mechanism. Angiogenesis 4(1):61-70, 2001.

11. Issued Patent: "Treatment of Chronic Inflammatory Diseases with CM101/GBS Toxin" C.G. Hellerqvist and B.D. Wamil: US Patent No. 6,028,060 - Granted February 22, 2000.

12. Yakes F.M., Wamil B.D., Sun F., Carter C.E., Hellerqvist C.G.: CM101 treatment overrides tumor-induced immunoprivilege leading to apoptosis. Cancer Research, 60 (20):5740-5746, 2000.

13. Sundell H.W., Yan H-P., Wamil B.D., Carter C.E., Stahlman M.T. and Hellerqvist C.G.: Isolation and Identification of a Group B -β Hemolytic Streptococcal Toxin from Septic Infants. J. Pediatrics. 137 (3):338-344, 2000.

14. Issued Patent: "Facilitation of wound healing with CM101/GBS Toxin" Hellerqvist C.G., Neeman M., and Wamil B.D., Abramovich R.: US Patent No. 5,858,991- Granted January 12, 1999.

15. Yan H.-P., Carter C.E., Wang E., Page D.L., Washington K., Wamil B.D., Thurman G. D., and C.G. Hellerqvist.: Functional Studies on the Anti-Pathoangiogenic Properties of CM101. Angiogenesis 2(3):219-233, 1999.

16. Issued Patent: "Facilitation of repair of neural injury with CM101/GBS Toxin" Hellerqvist C.G., Wamil A.W., Wamil B.D: US Patent No. 5,981,508 - Granted November 9, 1999.

17. Wamil A.W., Wamil B.D., Hellerqvist C.G.: CM101-Mediated Recovery of Walking Ability in Adult Mice Paralyzed by Spinal Cord Injury. Proc. Nat'l Acad. Sci USA, 95:13188-13193, 1998.

18. Wamil B.D., Yakes F.M., Page D.L., Yan H-P, Carter C.E., Sundell H.W., Fu C. Bardhan S., Johnson D.H. Hellerqvist, C.G.: CM101: An anti-pathoangiogenic polysaccharide reduces immunoprivilege of tumors. Cancer Detection & Prevention Impact of Cancer Biotechnology on Diagnostic and Prognostic Indicators in Predictive Oncology and Therapy. 22, Supplement 1:S-123-S-124, 1998.

19. Venkov C., Vorobieff N., Wamil B., Sun F., Yakes M., and Hellerqvist C.: A Quantitative Assay Using Basement Membrane Extracts and Embryonic Stem Cells to Study the Effects of Estrogen on Developing Vascular Endothelium *In Vivo.* Proceeding of the 20th European Conference on Microcirculation Eds: P. H. Carpentier, E. Vicaut and J.-L. Guilmot, Pub: Monduzzi Editore. Pages 213-217, 1998.

20. Wamil B.D., Thurman G.B., DeVore R.F., Wakefield G., Johnson D.H. and C.G. Hellerqvist. : Soluble E-Selectin in Cancer Patients as a Marker of the Therapeutic Efficacy of CM1O1, a Tumor Inhibiting Agent, Evaluated in Phase I Clinical Trial. J. Cancer Res. Clin. Oncol. 123: 173-179, 1997.

21. DeVore R.F., Hellerqvist C.G., Wakefield G., Wamil B.D., Thurman G.B, Minton P.A., Sundell H.W., Yan H., Carter C., Wang Y.-F., York G.E., Zhang M., and D. H. Johnson.: A Phase I Study of the Antineovascularization Drug CM101. Clin. Cancer Res., 3: 365-372, 1997.

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23. Wamil B.D., Wamil A.W., Rogers B.J.: Effect of anti-epileptic drugs and calcium channel active compounds on human sperm fertilizability. Mol. Andrology., Vol. 4 (1, 2): 137-138, 1992.

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25. Venkov C., Vorobieff N., Yan H.-P., Wamil B., Page D., and C. Hellerqvist.: Effect of Estrogen and CM101 on Endothelial Progenitor Cell Derived Neovascularization. (Angiogenesis.1999)

26. Thurman G.B., Wamil B.D., Quinn T.E., Zhang M., and Hellerqvist C.G.: Enhanced IL-6 Response in Tumor-Bearing Mice Injected With CM101: A Carbohydrate Isolated From Group B Streptococcus. J. Cancer Res. 1999

27. Hellerqvist C.G., Thurman G.B., Wamil B.D., Sundell H.W., Minton P., Carter C.E, Yan -P., Zhang M.-H., Johnson D.H., DeVore R.F., B. Wakefield.: Cytokine Responses of Cancer Patients Treated with CM101 - A Polysaccharide Anti-Cancer Agent Isolated from Group B Streptococcus. Clin. Cancer Res. 1999